
Topics of French Electoral Manifestos (1973–1993) and their Temporal Evolution through the lens of NLP

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Abstract

This report studies the thematic content of French legislative *professions de foi* from the Archelec corpus, covering five legislative elections between 1973 and 1993, and asks how those topics vary across parties, political families, candidate occupations, and election years. I compare three topic models: a classical Latent Dirichlet Allocation (LDA) baseline, a Nonnegative Matrix Factorisation (NMF) on TF-IDF features, and the more recent BERTopic pipeline based on sentence embeddings. NMF yields the cleanest, most interpretable topics and is used as the reference for the cross-tabulations with metadata; BERTopic is used as a state-of-the-art comparison and to study temporal evolution. Beyond mean topic prevalence per group, I implement a shift-share decomposition that separates changes in topic prevalence into a *composition* effect (which families speak louder) and an *intensity* effect (each family shifting its own discourse). The results show clearly stratified discursive spaces by political family, a well-known emergence of immigration and national-preference vocabulary on the far-right side from 1986 onward, and a recomposition of the left around the early-1980s alternation.

1 Introduction

The Archelec corpus [Gaultier-Voituriez, 2022] brings together more than 20,000 digitised *professions de foi* – the candidate manifestos sent to voters before each election – of the French Vth Republic. Each document is paired with structured metadata exported from the Sciences Po Archelec explorer (election date, candidate gender, occupation, declared party support, etc.). This combination of medium-length political discourse with rich candidate-level metadata makes the corpus a natural setting for topic modelling.

I focus on the 4 bullets of the *Topic Modelling & Temporal Evolution* track of the project sheet: (i) the thematic propensity of parties for a given election, (ii) the topics on which a party is significantly over- or under-represented relative to the corpus average, (iii) a semantic mapping of the manifestos so that documents can be projected into a thematic space, and (iv) the correlation between dominant topics and candidate metadata such as profession. To these I add a *temporal evolution* component, which is implicit in the project title and which the BERTopic family of models makes natively available.

The analysis is built on three models that, taken together, span the practical history of topic modelling: LDA [Blei et al., 2003], NMF [Lee and Seung, 1999], and BERTopic [Grootendorst, 2022]. My goal is not to declare a winner among them, but to use their differences to interpret the corpus more cautiously: a topic that emerges only in one of the three families is, in my opinion, less trustworthy than one that emerges in all three.

The code and the figures of the report are organised in a Python package, `archelec_topics`, with one submodule per stage of the pipeline (loading, preprocessing, models, interpretation, visualisation, metadata-aware analyses, temporal-evolution analyses). The interface design of the modelling subpackage – one abstract base class and one concrete wrapper per model family with a common

API for top words, document-topic matrix and dominant topic – is inspired by the topic-modelling infrastructure I worked on during my internship at the European Central Bank in 2025, simplified but also enhanced to the present project (at the ECB I was working on 4 languages at the same time, but I was working only with keywords matching and LDA algorithms, not NMF and BERTopic).

2 Related work and state of the art

Probabilistic topic modelling started with Latent Semantic Indexing [Deerwester et al., 1990] and its probabilistic variant [Hofmann, 1999], before LDA became the standard reference [Blei et al., 2003].¹ The decisive step from pLSI to LDA is a shift in what is being modelled. pLSI defines a probability distribution over the words of each individual document in the training set, with one free parameter θ_d (topics distribution) per document and no mechanism linking documents to each other. LDA, by contrast, models the generative process of an entire *corpus*: every document is treated as a sample drawn from a shared population, with its topic mixture θ_d itself drawn from a common Dirichlet prior. There are 2 consequences. First, the parameter count no longer grows linearly with the number of documents, since the θ_d are now latent variables governed by a small set of hyperparameters rather than free parameters of their own. Second — and crucially for any temporal or comparative analysis like the one I conduct in this report — the model can assign a topic distribution to a previously unseen document by projecting it onto the same latent simplex, something pLSI cannot do without retraining. In terms of estimation, the idea is to reverse-engineer the corpus generative process, or better said infer the model that could have generated such a corpus (thanks to Bayes law). It is most often performed through collapsed Gibbs sampling or variational Bayes; the Gensim implementation I rely on uses online variational Bayes [Hoffman et al., 2010].

From a different stream of research, NMF [Lee and Seung, 1999] provides a non-probabilistic alternative: it factorises a TF-IDF matrix as the product of two nonnegative matrices, a constraint that mechanically yields sparser, more “part-based” representations. The intuition, illustrate on faces in the original paper, is that since basis vectors can only be combined *additively*, each base must capture a localised, recognisable part rather than a global pattern that cancels out with another. Translated to text, this pushes each word toward loading strongly on one topic instead of being shared across many — without imposing orthogonality, sparsity emerges as a side-effect of the non-negativity constraint. As a consequence, on short to medium-length text, NMF often produces topics that are more readable than LDA at the same K [O’Callaghan et al., 2015].

However, both LDA and NMF treat documents as bags of words and ignore semantic similarity between vocabulary terms: synonyms (*ouvrier / travailleur / salarié*) are seen as unrelated tokens, polysemy is not modeled, and two manifestos that discuss the same topic in different words have nearly disjoint feature vectors. The emergence of *embeddings*, an alternative way of representing textual data as dense numerical vectors, has contributed to solving these issues of the bags-of-words representation. Instead of one dimension per vocabulary word, producing a sparse and non-semantic representation, embeddings project tokens, sentences and entire documents into a shared low-dimensional space in which geometric proximity reflects semantic similarity, so that two texts addressing the working class end up close to each other even when their bag-of-words signatures barely overlap. Successive generations of embeddings — from static word vectors (word2vec, GloVe) to contextualised embeddings produced by large pre-trained transformer models like Sentence-BERT [Reimers and Gurevych, 2019] — have progressively refined this idea.

BERTopic [Grootendorst, 2022], which is the modern endpoint of the topic-modelling timeline reproduced in the course slides, replaces the bag-of-words assumption by Sentence-BERT embeddings, projects them to low dimension with UMAP², clusters them with HDBSCAN³, and re-extracts topic words via a class-based TF-IDF (c-TF-IDF) on the resulting clusters. Other contemporary embedding-based approaches include Top2Vec [Angelov, 2020] and the Embedded Topic Model [Dieng et al., 2020].

¹I also read the very accessible overview written by the LDA co-authors Blei [2012], and consulted the recent survey Churchill and Singh [2022] for a comprehensive history of the field.

²[McInnes et al., 2018] UMAP is a non-linear dimension reduction technique able to preserve curved manifolds of the embedding space and both local and global structure of the data

³[Campello et al., 2013] HDBSCAN is a hierarchical density-based clustering algorithm that, unlike K-means, does not require the number of clusters to be specified in advance and flags low-density documents as outliers

A second family of topic models incorporates time *natively*: they treat the timestamp of each document as an observed variable that participates in the generative process rather than as an exogenous label used after fitting. One of the first models to do this is Dynamic Topic Model of Blei and Lafferty [2006], in which the corpus is sliced into time bins and the topic-word distributions of two consecutive bins are linked by a Gaussian random walk, forcing a smooth drift of each topic across periods. Wang et al. [2008] extend this idea to continuous time via Brownian motion on the topic parameters, and Dieng et al. [2019] combine it with the embedding-space view of ETM in their Dynamic Embedded Topic Model. Across this family, the common feature is that the temporal structure constrains the model itself, not just its post-hoc analysis, in contrast to the static topic models discussed earlier.

Concerning French electoral manifestos specifically, the literature has so far mostly relied on hand-coded categorisations such as the Manifesto Project [Lehmann et al., 2023] or on lexicometric tools dedicated to French (Iramuteq, Lexico). A few recent works apply LDA to French political discourse [Bourgeois et al., 2022], but, to the best of my knowledge, the specific Archelec OCR’ed corpus has not yet been studied with embedding-based topic models. Hence the following analysis.

3 Data

3.1 Corpus and join with metadata

The corpus is built from two heterogeneous sources: an OCR’ed text file per *profession de foi* from the Arkindex/Teklia repository, and a CSV of metadata exported from the Archelec Sciences Po explorer. I join the two on the document identifier with a custom loader. After the join, 21,697 manifestos are available; 530 transcriptions ($\sim 2.4\%$) have no metadata counterpart (these are in fact *bulletins de vote*, not *professions de foi*, and I drop them). The remaining 21,167 documents cover five legislative elections (1973, 1978, 1981, 1988, 1993), $\sim 78\%$ of them being first-round manifestos. A small date anomaly in 1978 (three rounds rather than two) is fixed manually, and “*non mentionné*” strings are converted to true missing values.

3.2 Statistical description

Figure 1 gives a first description of the corpus that will guide the rest of the analysis. The volume of manifestos varies between roughly 3,100 (1981) and 5,800 (1993), so per-year averages should be read with these unequal sample sizes in mind. The political-family panel is computed on the harmonised *titulaire-soutien* field after I collapsed the 1,523 raw labels (which include alliances such as “*RPR;UDF*”) into a curated mapping of canonical parties, themselves grouped into eight political families plus a residual category. The landscape that comes out is the expected one: a stable left/right backbone, with some expected movements within each large political families, (i) such as within the big center-right-extreme-right family, the hegemony of the center and traditional right that leaves more and more place to the extreme right, which end up on par with the center and the traditional right at the end of the studied period; or (ii) such as the almost disappearance of the “Gauche alternative” and the strong decline in socialists candidacies during Mitterrand’s first presidential mandate. On top of that, we can clearly see the appearance of the Front National from the 1980s onward, and the late emergence of organised political ecology between 1988 and 1993.

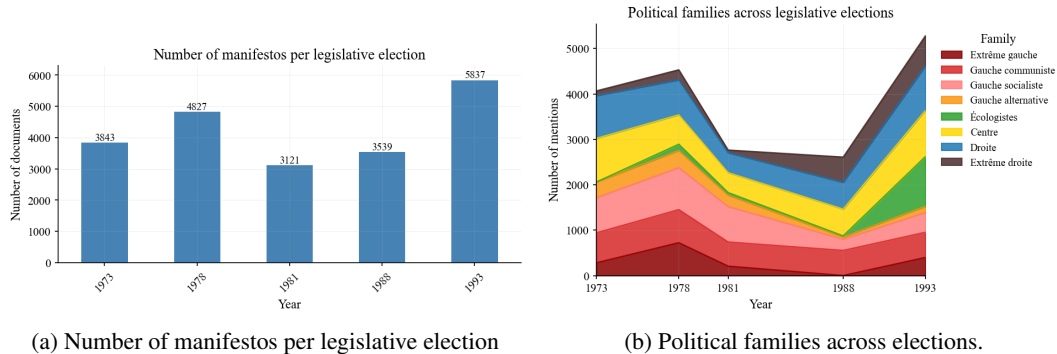


Figure 1: Descriptive view of the corpus before any modelling

Coverage of the candidate metadata is uneven. Sex is reliably recorded ($\sim 98\%$, with $\sim 84\%$ men, $\sim 10\%$ women): as expected female candidacies remain a small minority across the five elections (see further analysis in Appendix A.1). Party support is missing for $\sim 18\%$ of candidates; profession is missing for $\sim 44\%$, which restricts any topic $\times$ profession analysis to roughly half of the corpus. I therefore treat sex as a secondary covariate and concentrate on parties, political families, occupations, and occupational categories.

The study of the most represented professions (and categories of professions) among the candidates (see Appendix A.2 for further analysis), and a cross-tabulation of the top-10 parties and the top-10 occupations (see Appendix A.3) confirms patterns that are well known sociologically: liberal professions and teaching dominate among candidates, the PCF concentrates a substantially higher share of *ouvriers* than any other party, while the right-of-centre formations recruit more disproportionately from *cadres* and *professions libérales*. This descriptive layer matters for the topic analysis because we know that some topics will be picked up partly through the lens of the candidate's occupation and not only through their party.

3.3 Preprocessing

The text comes from OCR and is, in places, quite noisy. Following the recipe from the course slides, my pipeline (in `archelec_topics.preprocessing`) is the following:

1. **German paragraph filter** – a non-negligible share of the Alsace manifestos are bilingual; I drop paragraphs with a German common-word ratio above the threshold (10%), which is conservative enough to keep monolingual French text intact.
2. **Acronym expansion** – before lowercasing, two-letter acronyms such as FN, LO, RPR, UDF, CDS are expanded to their full underscore-joined form (*front_national*, *lutte_ouvriere*, ...) to enhance readability and because, without this, FN and LO would be filtered out by the minimum-token-length rule.
3. **Compound expressions** – a hand-curated list of ~ 200 multi-word political expressions (*programme_communiste*, *union_de_la_gauche*, *droits_de_l'homme*, *front_populaire*, ...) is glued together with underscores *before* spaCy is called, so that they survive as single tokens and can be handled uniformly afterwards. This is a way of taking into account n-grams without doing a very heavy computation of all the n-grams in the corpus (that is already very heavy for bigrams). The limitation of this method is that I impose the n-grams to look at, and I don't let the data tell me the interesting n-grams. But with a bit of political historical knowledge of the period, I managed to spot interesting n-grams.
4. **OCR cleanup** – recurring archival headers (*Sciences Po / Fonds Cevipof*, *République française*, *Élections législatives*) are removed, line breaks with hyphen (–) as well as hyphen inside words (OCR errors) are mended to get back to full words for the future tokenisation, and non-letter characters are stripped.
5. **Tokenisation, lemmatisation and POS filtering** via spaCy (`fr_core_news_md`); only nouns, proper nouns, adjectives and verbs are kept.
6. **Stopword removal** on the lemmatised, accent-stripped tokens, using spaCy's default French stopword list augmented with a domain-specific list (electoral vocabulary, formulas of address, month names, German leftovers, common candidate first names, OCR garbage). This list was refined iteratively after several runs of LDA.

This iterative aspect is important: I started with a much shorter stopword list, ran LDA, observed which junk tokens were polluting the topics, fixed them, and ran again. From my previous experience with LDA at the European Central Bank on a multilingual press corpus, I know that this loop of preprocessing is an enormous determinant of the final readability of the topics. Two diagnostic functions (`summarize_token_frequencies` and `summarize_compound_hits`) help drive this loop quantitatively.

After preprocessing I apply a quality filter: I keep documents with at least 500 raw characters (a clear OCR-failure floor), at least 50 useful tokens (below this LDA simply does not have enough signal to assign a stable distribution), and at most 1,000 useful tokens (which doesn't drop much documents). The exact thresholds were chosen by inspecting the bimodal left tail of the length histogram.

4 Models choice and its justification

The related-work section above lists at least seven topic-modelling families that I could have applied to this corpus: LDA, NMF, BERTopic, Top2Vec, ETM, DTM (and its continuous-time variant), and D-ETM. I retain three – LDA, NMF and BERTopic – and apply a post-hoc temporal analysis to all three rather than using a natively dynamic model. This section explains both choices.

Why these three models in particular. The deepest reason is that LDA, NMF and BERTopic span three genuinely different families of inductive biases: probabilistic bag-of-words, algebraic bag-of-words, and embedding-based clustering. A topic that emerges only in one of the three is, in my opinion, less trustworthy than one that emerges in all three (Figure 5, topic alignment accross methods is precisely the diagnostic that exploits this triangulation). Adding Top2Vec would have added a fourth model in the same family as BERTopic (embedding + clustering), bringing little methodological diversity. ETM would have been more orthogonal (it is probabilistic and embedding-based) and is, in retrospect, the model I regret the most not having included. There is also a more pragmatic reason: I came to this project with prior experience on LDA from an internship at the European Central Bank, and I wanted to anchor the analysis on a model whose hyperparameters and output objects I fully understood, before adding a state-of-the-art layer. NMF was the natural algebraic counterpart, both because of its pedagogical clarity and because the literature reports it to produce more readable topics than LDA on short to medium texts, which the manifestos qualify as. BERTopic provided the modern state-of-the-art embedding-based comparison.

Why a post-hoc temporal analysis rather than a natively dynamic model. The case for a dynamic topic model rests on the assumption that the *vocabulary itself* of a given topic drifts continuously over time, and that this within-topic drift is what one wants to recover. On the twenty-year window of the Archelec subset, with five legislative elections, this is not the dominant phenomenon I want to study. The political vocabulary of 1973 and 1993 is broadly the same – the same words for unemployment, security, Europe, ecology, immigration – and what changes is much more the *prevalence* of each theme and *which actors carry it* than the words themselves. In that regime, the right object is a stable, time-invariant set of topics whose trajectory across periods can be tracked, which is exactly what BERTopic gives me out of the box.

A second, methodological reason matters even more, in my opinion. The political history of the 1973–1993 period is punctuated by structural ruptures – the 1981 *alternance*, the institutional rise of the Front National after 1984, the emergence of organised political ecology in 1988–1993. These discontinuities are not noise to be smoothed out: they are precisely the phenomena I wanted to try to detect. The Gaussian random walk prior at the heart of DTM, cDTM and D-ETM regularises consecutive time slices toward each other, which is the right inductive bias for slow lexical drift in a stationary corpus (a century of *Science* articles, in the original Blei–Lafferty experiment) but the wrong one for a corpus in which the interesting structure is brusque changes in prevalence. A post-hoc analysis that recomputes per-period statistics on a fixed set of topics, by contrast, leaves the trajectory of each topic fully free.

The third reason is comparability across models. Applying the same per-period aggregation to LDA, NMF and BERTopic makes their outputs structurally comparable, and lets me run the cross-model diagnostics (Jaccard alignment, shift-share decomposition, Jensen–Shannon distance between successive elections) on a uniform basis. Replacing BERTopic by DTM or D-ETM would have produced output objects that cannot be aligned cell-for-cell with the static models. A fourth, more pragmatic reason is that the Gensim wrapper of DTM is deprecated and D-ETM has no maintained pip-installable implementation; this is admittedly the weakest of the four arguments, but it is honest.

5 Experiments and qualitative evaluation

5.1 Hyperparameter finetuning, or choosing the number of topics K

Sweep analysis for LDA and NMF. I performed a series of sweeps for $K \in \{5, 8, 10, 12, 15, 18, 20, 22, 25, 28, 30\}$ for both LDA and NMF, evaluating the C_v coherence [Röder et al., 2015] and the topic diversity [Dieng et al., 2020]; results are reproduced in Figure 2. Three observations come out, which I think are informative beyond the simple choice of K .

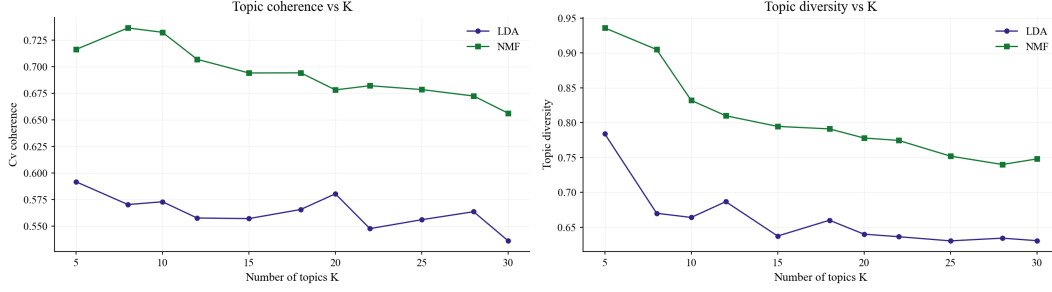


Figure 2: Coherence (C_v , left) and topic diversity (right) of LDA and NMF as a function of K . NMF dominates LDA on both metrics for the same K , and both metrics flatten between $K=8$ and $K=12$.

First, NMF dominates LDA on both metrics for every K I tested. This is in line with the literature on short-to-medium documents [O’Callaghan et al., 2015] and consistent with what I know of the two models: NMF’s sparsity-inducing constraint produces cleaner topics on TF-IDF-weighted texts, while LDA’s Dirichlet smoothing distributes mass over more words and makes the topics noisier. Second, both metrics are quasi-monotonically decreasing past a plateau located around $K \approx 8\text{--}12$, except LDA that increases again around $K \approx 18\text{--}20$, which suggests both (i) that the corpus has a relatively low number of truly distinct themes and (ii) that LDA and NMF might actually need very different number of topics to perform the best (in their own range of ‘best’). Third, none of the metrics is very stable in the sense that they don’t reward the same K (both across methods and within a method) – coherence peaks at $K=8$ for NMF but at $K=20$ for LDA (which is actually very close to $K=5$), diversity peaks at $K=5$ for both method, but if we ignore this very low number of topics, it peaks at $K=8$ for NMF and at $K=12$ for LDA (increasing again around $K \approx 18\text{--}20$).

A purely metric-driven choice would give $K=8$. After labelling the topics by hand at $K \in \{8, 9, 10\}$ I converged on $K=9$ as the better compromise. At $K=8$ the model collapses environmental and economic-development vocabulary into a single hybrid topic, which I want to keep separate; and at $K=10$ the *Union de la gauche* topic of NMF splits into two near-duplicates that say essentially the same thing. At $K=9$ a substantively meaningful topic appears in NMF, namely an *autogestion* / anti-capitalist register (PSU, ouvriers, capitalisme) that is absorbed into the broader class-struggle topic at $K=8$. This distinction matters because it splits the left into three different registers (institutional, economist-radical, alternative) which I expected to map onto distinct political families.

I think that the idea that LDA might need more topics than NMF to be readable is linked to my choice of priors. With the option ‘auto’ on alpha and eta, the two concentration parameters of the Dirichlet distributions, Gensim learns the two Dirichlet hyperparameters from the data, allowing the model to give topics asymmetric prior weights, i.e. to recognise that some topics are intrinsically more frequent than others, which Wallach et al. [2009] argue is more robust than the default symmetric prior on heterogeneous corpora (hence my choice). However, this optimisation does not push them as low as one could manually fix them: a hand-tuned $\eta = 0.01$, for instance, would put the prior mass much closer to the corners of the simplex and yield peakier word distribution for each topic, mimicking NMF’s sparsity. Lowering α , similarly, encourages each document to load on only a few topics. A grid search over $(\alpha, \eta) \in \{0.01, 0.1, 0.5\}^2$ at fixed K is, in my opinion, the natural next step to narrow the gap with NMF – no time to run it, and NMF was already giving me good topics.

Good embeddings for BERTopic. The reasoning above applies to LDA and NMF, that both require K to be fixed before fitting. BERTopic works very differently. Its clustering backbone is HDBSCAN, which finds clusters of variable density and outputs a number of topics that depends on the data, not on a target. I can either force a target number (`nr_topics=k`, in which case BERTopic merges similar clusters at the end of the fit until reaching k) or leave it at “auto” and let HDBSCAN decide. Although it is expected from BERTopic when fixing a target number of topics to separate very niche topics from a massive aggregative topic, my first run using `nr_topics=15`, for direct comparability with LDA/NMF, produced a pathological output: a single catch-all topic absorbing $\sim 50\%$ of the corpus, plus several near-duplicate clusters for the nationally templated campaigns of small parties (LO 1993, FN 1988, *Nouveaux écologistes* 1993). After investigation, I realised that the SentenceTransformer model I was using (paraphrase-multilingual-mpnet-base-v2) ships with a default `max_seq_length` of 128 tokens, while the WordPiece-tokenised median of my corpus

is at 1089 tokens (P90 = 1955). This means **100% of my manifestos were silently truncated** by the tokenizer, with only the first $\sim 12\%$ of each document reaching the transformer. The clusters were therefore reflecting administrative headers (candidate name, constituency label, “Sciences Po / fonds CEVIPOF” boilerplate, OCR artefacts in the first hundred words), not programmatic content. I re-embedded the entire corpus with BAAI/bge-m3 configured at `max_seq_length = 2048` – a long-context multilingual model from 2024 that supports up to 8192 tokens architecturally, and at 2048 covers $\sim 95\%$ of my manifestos in their entirety. The recomputation took approximately two hours on a Colab T4 GPU.

With these new embeddings and `n_topics="auto"`, HDBSCAN finds 68 natural topics with 44% outliers. The high outlier rate is not a bug – it is the expected behaviour of HDBSCAN on a heterogeneous corpus, where many idiosyncratic manifestos do not lie in any clear density region and the algorithm refuses to assign them rather than forcing them into a nearest-cluster label (which is what LDA’s soft assignment implicitly does). Forcing `n_topics` at fit time would re-trigger the catch-all behaviour I observed on the first run. Instead, I reduce the output *a posteriori* in two steps. First, I re-assign outliers via embedding similarity (`reduce_outliers(strategy="embeddings")`), which moves each outlier to the topic whose centroid is closest in bge-m3 embedding space; this is consistent with the criterion that drove the original clustering. Second, I merge topics hierarchically (`reduce_topics(nr_topics=25)`) until reaching 25 topics, by iteratively fusing the pair with the highest c-TF-IDF cosine similarity. The target of 25 is roughly $K_{\text{LDA}} + 15$, with the margin intended to preserve the additional fine structure that BERTopic legitimately recovers (regional clusters, ideological micro-currents). (See the hierarchical clustering in Appendix ??, figure ??). I think this two-step approach is a fair compromise: I do not impose a target at the moment when it would distort the clustering, but I do align the output’s granularity to make cross-model comparison readable.

5.2 Topic interpretation and labelling

Topic models return distributions over the vocabulary, not semantic categories. The labelling step is therefore subjective and I follow the standard recipe: read the top-12 words of each topic, then read the three manifestos with the highest weight on that topic, and pick a label that fits both.

Figure 3 shows the NMF top words: they fall into clear semantic registers, which I label as: (0) *Action locale & développement économique*, (1) *Anti-réformisme révolutionnaire*, (2) *Cause animale & nature*, (3) *Préférence nationale & baisse d’impôts*, (4) *Insécurité et immigration*, (5) *Écologie politique*, (6) *Lutte des classes*, (7) *Union de la gauche and Programme commun*, (8) *Autogestion & anticapitalisme*. Only the topic (1) was hard to label cleanly, which I see as some residual noise (mentions of national figures and electoral generalities); I keep it in the analysis but flag it.

The LDA topics at $K=9$ are harder to label (see Appendix C, Figure 15). Topic 2 mixes school vocabulary, regional anchors and a Front National mention without an obvious unifying theme; Topic 3 combines opposition rhetoric with economic-crisis vocabulary in a way that could capture either a left-wing critique of austerity or a right-wing campaign on unemployment. These ambiguities reflect LDA’s tendency to produce smoother, more overlapping distributions, which is informative in itself but more difficult to label.

The BERTopic model, after the two-step reduction described in Section 5.1, returns 25 topics. I labelled them by reading the top-12 c-TF-IDF words and the three most representative documents of each cluster, like for LDA and NMF (see Appendix C, Figure 16). Two things are worth noting about this labelling. First, many topics are dominated by vocabulary that could intuitively be associated with only one party (FN, LO, PSU, Nouveaux écologistes), or even periods of certain parties. I deliberately chose *thematic* labels because the shift-share decomposition that follows requires labels that can in principle be carried by multiple families: I am precisely trying to understand if the semantics that one would expect to be used by this or that political party are really the ones picked up by the model, and how the semantics can actually evolve within a political family. Second, BERTopic recovers a few topics that LDA and NMF miss, especially territorial ones (Alsace, Bretagne, Corse, DOM-TOM) and ideological micro-currents (PSU autogestionnaire, LCR vs Parti des travailleurs lambertiste, CNI Pinay). A contrario, it also (still) produces one large catch-all cluster (Topic 0, *Mandat sortant & confirmation de confiance*) that captures the generic boilerplate of incumbent and moderate manifestos, as well as common electoral vocabulary that wasn’t filtered out – this cluster is structurally large and stable across elections, and I treat it accordingly in the temporal analysis below.

NMF — Word clouds

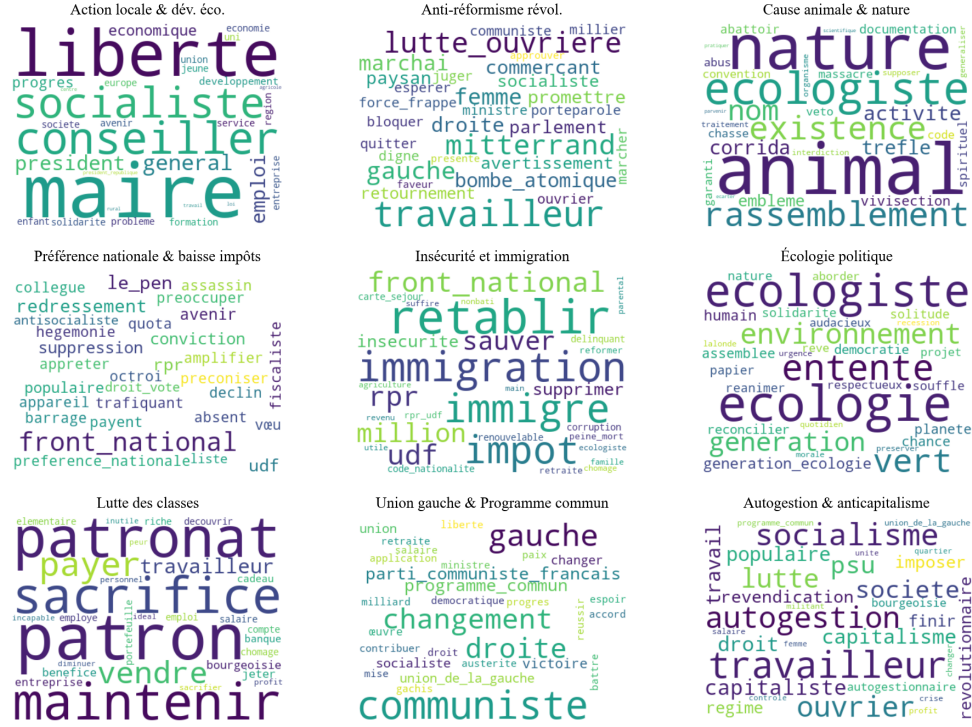


Figure 3: NMF – word clouds (font size proportional to NMF component weight). Labels are assigned manually after reading the top words and the three most representative documents per topic.

5.3 Cross-model alignment

Figure 4 shows the Jaccard similarity between the top-15 words of the LDA and NMF topics on the right. The diagonal-like alignment confirms that LDA and NMF recover essentially the same backbone of topics, with one-to-one matches for political ecology, class struggle, immigration / national preference, and Union de la gauche. The few mismatches are exactly the LDA topics I struggled to label.

The LDA $\leftrightarrow$ BERTopic alignment 5 on the right is much sparser and – I think – more interesting to read. Because BERTopic has 25 topics against LDA’s 9, several BERTopic topics map to the same LDA topic (this is granularity, not contradiction): for instance the LDA *Insecurity and immigration* topic aligns with three BERTopic topics, separating the FN’s 1988 *Appel aux Français* register from its 1993 *Constat de crise* register and from a more focused security-and-apprenticeship sub-topic.

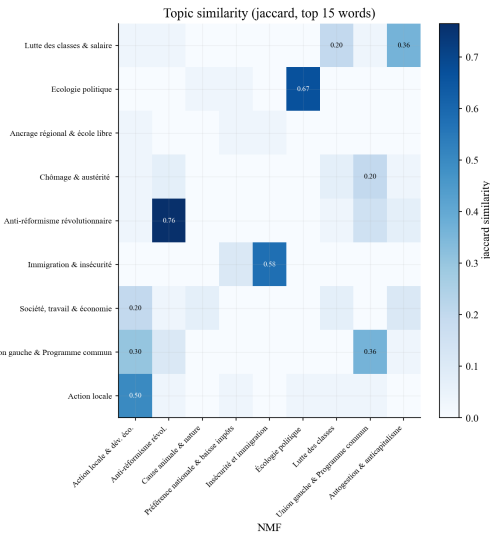


Figure 4: LDA-NMF cross-model alignment

The LDA *Class struggle* topic similarly splits into two BERTopic sub-topics that distinguish the 1973–1978 *témoignage militant* register from the 1993 *patrons & chômage* register of Lutte Ouvrière. The fact that the same backbone emerges across three very different models is, I think, the strongest argument for taking the topics seriously, and the additional BERTopic granularity is a useful complement rather than a contradiction.

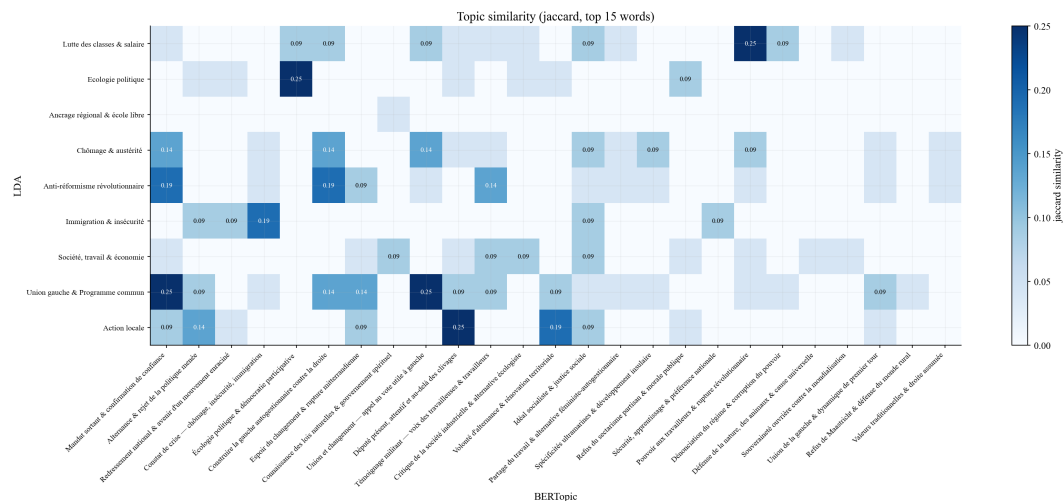


Figure 5: Cross-model alignment of topics by Jaccard similarity on the top-15 words.

6 Results: topics by parties & professions, evolving across time

As explained before, the topics generated by NMF and BERTopic are much easily readable (than the messy LDA ones), so that I use both models for the analysis. For the main metadata-driven analyses I use BERTopic as the reference model, but you can find a similar analyses with NMF topics in the Appendix D and comparison with BERTopic findings. The reason is that BERTopic's finer granularity (25 topics against 9) lets me read more nuanced patterns at the political- family level.

6.1 Topics by political family

Figure 6 shows the per-topic z-score of each political family relative to the corpus average (see Appendix D, Figure 17 for the NMF version). The z-score normalisation makes over- and under-representation symmetric and visually readable. As before, a coalition document contributes to every family supported by its candidate.

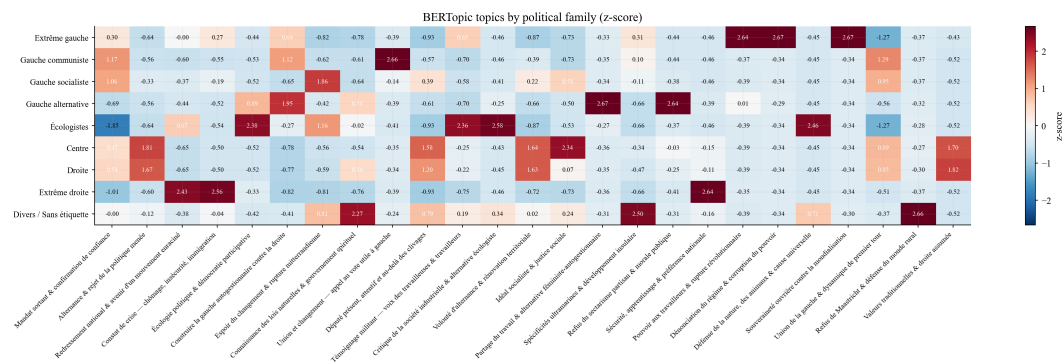


Figure 6: BERTopic topics by political family, expressed as a per-topic z-score of each family relative to the corpus average. The z-score reads as: *among the documents of this political family, the average weight of this topic is higher (red) or lower (blue) than the average topic weight over all political families.*

The picture confirms the standard sociology of French political discourse and refines a few patterns that NMF aggregated. *Pouvoir aux travailleurs & rupture révolutionnaire* and *Témoignage militant* are strongly over-represented on the *Extrême gauche*, with the two registers actually splitting between the LCR/PSU side (revolutionary) and the LO side (testimonial). *Union de la gauche* and *Idéal socialiste & justice sociale* are the signature of the *Gauche communiste* and *Gauche socialiste* families, with a clean separation between the two that NMF was collapsing. *Écologie politique* and *Défense de la nature* concentrate on the *Écologistes*, with a secondary loading on the *Gauche*

alternative (PSU autogestionnaire registers). On the right side, *Constat de crise – chômage, insécurité, immigration, Redressement national* and *Sécurité, apprentissage & préférence nationale* are the three signatures of the *Extrême droite*, and again the three-way split is informative – the FN does not speak in one register but in three, depending on year and constituency. The moderate *Droite* and *Centre* families load mostly on *Mandat sortant & confirmation de confiance* and *Action locale*, which is consistent with their incumbent position and with the flatter signature observed in NMF. The equivalent NMF heatmap’s backbone agrees with BERTopic but with much less granularity.

An expected observation is that the specialisation of the discourse is strongest at the radical-left and far-right ends, weakest in the centre and moderate right. BERTopic adds that, within the radical fringes, the specialisation is not on a single topic but on a small bundle of register variants – the FN runs three sub-registers, the radical left runs at least three (testimonial, revolutionary, autogestionnaire), and each is carried by an organisationally distinct sub-family (see Appendix E, Figure 21 to further study the party specialisation of the discourse).

6.2 Topics by occupation

Crossing topics with profession metadata gives a much weaker signal than the political family-level analysis, mostly because $\sim 44\%$ of profession data is missing. The two patterns I noted in the NMF version (Appendix D Figure 19 & 20) – *Justice & droit* loading on the immigration register, and *Enseignement et recherche* loading on the self-management/anti-capitalism register – are also present within BERTopic topics (Appendix E Figure 22 & 23), but I do not give this analysis the same weight as the political affiliations one.

6.3 Temporal evolution: prevalence over elections

Figure 7 shows the BERTopic prevalence per year, with the full picture on the left (top 8 topics including the catch-all) and a zoomed-in version on the right (same top 8 minus Topic 0). I plot both because the full picture is dominated by Topic 0, which compresses the dynamics of all other topics under 0.20, while the zoom reveals what is actually happening in the rest of the corpus.

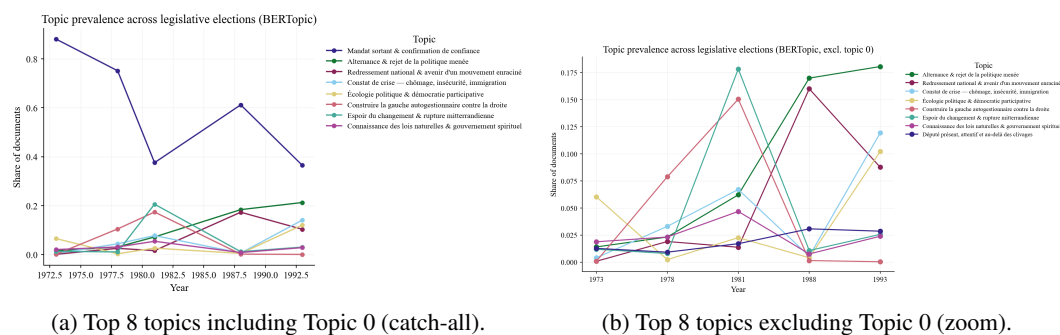


Figure 7: BERTopic normalized topic prevalence across the five legislative elections. Colour assignment is fixed across both panels: Topic 0 occupies the dark-blue slot in (a) and is removed in (b).

The shape of Topic 0 itself deserves a comment, because it is a feature of how BERTopic clusters work and not just a noise topic to discard. Topic 0 absorbs around 50% of the corpus and corresponds to the generic vocabulary that almost every candidate uses – mandate, local service, republican values, confidence renewed, etc. – amplified by the `reduce_outliers` step, which redistributed the HDBSCAN outliers toward their nearest centroid in embedding space, and in many cases that nearest centroid was Topic 0 because outliers tend to be precisely the documents that mix several registers without committing to any one. Its slow decline from ~ 0.88 in 1973 to ~ 0.36 in 1993 is therefore not a political phenomenon by itself; it reflects the diversification of the corpus into more clearly identifiable registers over time, as new families (FN, écologistes) bring in clearly distinct vocabularies that pull documents away from the generic cluster. I think this is a useful illustration of the structural difference between BERTopic and the bag-of-words models: LDA and NMF distribute the boilerplate vocabulary across all topics with low weights, while BERTopic isolates it into a single cluster.

On the right, the zoomed panel shows the substantively interesting dynamics, and they match French political history rather closely. *Construire la gauche autogestionnaire contre la droite* and *Espoir du changement & rupture mitterrandienne* both peak in 1981 – the two registers of the left-wing victory of 1981 – and decline noticeably afterwards. The *Constat de crise – chômage, insécurité, immigration* topic appears from 1988 onward, mirroring the institutional rise of the FN after the 1984 European elections, and grows again in 1993. *Alternance & rejet de la politique menée* rises monotonically from 1981 to 1993, which is a register that emerges in opposition discourses across the period, picked up by both the right (1986 cohabitation, 1993 alternance) and the FN. *Écologie politique & démocratie participative* appears in 1973 (collectif Écologie 78, écologistes indépendants) and re-appears in 1993 with Génération Écologie and les Verts, with a gap in 1981-1988. Read together, I think these trajectories show that the corpus is rich enough to detect first-order political shifts in time, and that the topic-modelling layer does what it is supposed to do. The LDA and NMF temporal evolution show similar results, but again with less granularity.

6.4 Decomposing the temporal change: composition vs intensity

Mean prevalence per year mixes two very different kinds of change. A topic can become more prevalent either because the families that always carried it gain documents (*composition* effect: “who speaks?”) or because the families themselves shift their own discourse toward that topic (*intensity* effect: “what each family says”). To separate the two I implement an Esteban-style symmetric shift-share decomposition. Let w_{tf} be the share of family f in the documents of year t , and P_{tfk} the mean weight of topic k in the documents of (t, f) . The corpus-level prevalence of topic k in year t is $P_k^{global} = \sum_f w_{tf} P_{tfk}$, and the change between two elections decomposes as

$$\Delta P_k^{global} = \underbrace{\sum_f \Delta w_f \bar{P}_{fk}}_{\text{composition}} + \underbrace{\sum_f \bar{w}_f \Delta P_{fk}}_{\text{intensity}},$$

where $\bar{w}$ and $\bar{P}$ are the means of the two end-points.

I first computed the decomposition over the full 1973–1993 window. This is informative for the global picture but loses the chronology – the post-1981 collapse of *Programme commun* rhetoric is not visible if you only look at the start and the end. I thought it more interesting to look at the decomposition for each consecutive transition (1973→1978, 1978→1981, 1981→1988, 1988→1993) and plot the four panels side by side in Figure 8. For BERTopic I keep the 9 topics with the largest cumulative $|\Delta|$ across the four transitions, so that the same 9 topics appear on each panel and the reader can read a topic’s trajectory horizontally. The same plots for LDA and NMF, as well as the global 1973–1993 decomposition for the three models, are reproduced in Appendix G.

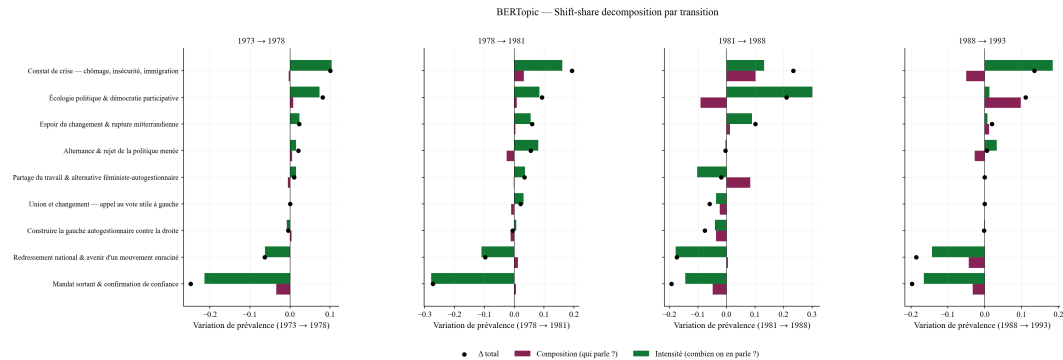


Figure 8: BERTopic – shift-share decomposition by consecutive election transition, restricted to the 9 topics with the largest cumulative variation. *Composition* corresponds to “who speaks more / less” and *intensity* to “each family changes its own discourse”. Black dots indicate the total Δ .

The four panels read together tell, I think, the most interesting story of this analysis. Three patterns stand out.

First, the rise of the *Constat de crise – chômage, insécurité, immigration* register between 1981 and 1993 is, in the early phase (1981→1988), a pure composition effect – the FN appears as a measurable share of the corpus and brings its own signature register. But in the second phase (1988→1993), the intensity component becomes positive too: the same register is picked up by other families. This is consistent with the political-science literature on the diffusion of FN-coined frames into mainstream right and even centrist discourse during the Pasqua–Méhaignerie cohabitation, and I think it is the cleanest example of why the composition–intensity separation is worth doing – a global decomposition over 1973–1993 would have shown a mixed effect and missed the chronology.

Second, the collapse of the *Construire la gauche autogestionnaire* and *Espoir du changement* registers is concentrated in the 1981→1988 transition and is dominated by an *intensity* effect. The PCF and the PS are still present in 1988, but they have shifted away from these two registers – the prophetic register of the opposition is no longer functional after the alternance, and the autogestionnaire register loses its carrier (the PSU dissolves in 1989) but also stops being adopted by neighbouring families. I think this is the post-1983 *tournant de la rigueur* effect made quantitatively visible.

Third, *Alternance & rejet de la politique menée* grows gradually across all four transitions, and the growth is split between composition and intensity in different ways depending on the period. This register seems to be a structural one in opposition rhetoric: as the carriers of opposition change (right in 1981–1986, left in 1988–1993, FN throughout), the same register is reused with adapted content. I think it is the topic that most clearly illustrates how BERTopic’s thematic labels capture a register that travels across parties.

These three readings would not be possible with a single 1973–1993 decomposition, which is why I now consider the per-transition version the central piece of the analysis.

6.5 Amplitude of the recomposition: Jensen-Shannon distance

As a complementary, single-number measure, I compute the Jensen-Shannon distance between the corpus-level topic distributions of two successive elections. Figure 9 reports the four transitions for the three models.

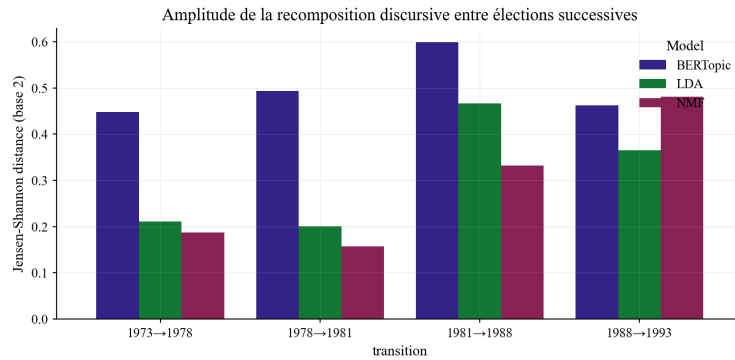


Figure 9: Jensen-Shannon distance between the corpus-level topic distributions of successive elections, for the three models.

Two observations. The absolute level of the BERTopic curve is systematically above that of LDA and NMF; this is structural, because BERTopic’s hard cluster assignments produce more peaked aggregated distributions, and JSD is mechanically more sensitive to displacements between peaked distributions than between flat ones. So the absolute BERTopic JSD should not be compared to the absolute LDA or NMF JSD. What is comparable is the *ranking of the four transitions* within each model. All three models concur on the same ordering: 1981→1988 is the largest discursive recomposition, followed by 1988→1993, with 1973→1978 and 1978→1981 markedly smaller. The fact that the most disruptive transition is 1981→1988 rather than 1978→1981 (Mitterrand’s election) is, I think, a substantive result: the rhetorical recomposition of the corpus does not synchronise with the political alternance, it lags by a cycle and unfolds during the 1986–1988 cohabitation – exactly the period when the PCF declines electorally, the FN consolidates, and the PS abandons the *Programme commun* register. I find this convergent-across-models finding the most robust signal of the temporal analysis.

7 Discussion, limitations and future work

What I think the analysis shows. Within the limits of an OCR’ed corpus and of unsupervised topic modelling, I think my analysis supports five substantive conclusions.

1. *First*, the discursive space of the French legislative *professions de foi* between 1973 and 1993 is well stratified by political family, with the strongest specialisation on the radical-left and far-right ends and a flatter signature in the centre and the moderate right.
2. *Second*, the radical fringes do not speak in a single register but in a small bundle of organisationally distinct sub-registers that BERTopic recovers and bag-of-words models tend to collapse – three FN registers, three radical-left registers, two ecology registers.
3. *Third*, the post-1981 decline of the *Programme commun* / class-struggle vocabulary is, in this corpus, an intensity effect concentrated in the 1981→1988 transition: the historically left-wing families remain present, but they speak about other things, which is consistent with ...
4. ... the *fourth* key point that the temporal recomposition of the discursive space peaks in 1981→1988, not in 1978→1981, which is robust across the three models, happening thus after the "alternance" happened and not before.
5. *Fifth*, the rise of the crisis / immigration / insecurity register is a composition effect in 1981→1988 (the FN appears) and an intensity effect in 1988→1993 (the register diffuses to other families), also contributing to the peak of the political discourse recomposition between 1981 and 1988.

Limitations. I see several ways of improving my work. First of all, the document length drops noticeably between 1978 and 1988, and I cannot tell with certainty whether this is a real stylistic shift or a transcription-quality effect. A way to control for this would be to fit topic models on length-stratified subsamples; I did not have time to run the experiment. The second limitation, also related to the quality of the data, is the missing metadata – profession is missing for 44% of candidates, which restricts the topic $\times$ profession analysis. The third is that I only used C_v and topic diversity for model selection on LDA/NMF, while a richer evaluation would also include perplexity, reconstruction error, and ideally an external benchmark such as the Manifesto Project codebook Lehmann et al. [2023]. The fourth is that I did not tune the Dirichlet hyperparameters of LDA, which I believe is responsible for a significant share of the LDA / NMF gap. The fifth is that BERTopic returns hard cluster assignments, which mechanically inflates the JSD relative to the soft-assignment models; running `approximate_distribution` would harmonise the absolute levels at a non-trivial computational cost. The sixth, more specific to BERTopic, is the choice of the embedding model and of the granularity of the post-hoc reduction – I used `bge-m3` at 2048 tokens and reduced to 25 topics, but neither value is optimal in any well-defined sense, and a sensitivity analysis on these two choices would strengthen the conclusions.

Things I wish I had had time to do. On the modelling side, three extensions feel especially natural. The first is the Dynamic Topic Model of Blei and Lafferty [2006], which I considered seriously but had to drop because of the deprecated wrapper and the time to do the project. With five time slices and $\sim 21,000$ documents this would in principle let me track *within-topic* drift of the vocabulary, not only the prevalence of a fixed set of topics. The second is the Embedded Topic Model Dieng et al. [2020], which would combine LDA’s probabilistic structure with embedding-based semantics, and might in particular help close the LDA / NMF readability gap. The third, more aligned with the project, would be the Structural Topic Model Roberts et al. [2019], which uses the candidate metadata as a covariate in the topic model itself – this would let me test whether *the same family* actually shifts its discourse between 1973 and 1993, conditional on the year, instead of inferring it after the fact through shift-share. Finally, on the embedding side, comparing `bge-m3` to other long-context multilingual models on the same corpus would help quantify how much of the BERTopic signal comes from the embedding step itself.

8 Conclusion

I built a small Python package and a single notebook that take the Archelec *professions de foi* from raw OCR text to a substantive analysis of the discursive space of the French legislative elections between 1973 and 1993. Three topic models, picked to span the practical history of the field, give a remarkably consistent backbone of topics. NMF at $K=9$ is the cleanest of the three on bag-of-words metrics; BERTopic with bge-m3 embeddings at 2048 tokens, post-hoc reduced to 25 topics, brings a finer semantic granularity that is what I end up using for the metadata-driven and temporal analyses. Within this analysis, the combination of per-topic z-scores for each political family, in between elections shift-share decomposition and Jensen-Shannon distance gives a quantitative way to separate two very different mechanisms of discursive change – composition (“who speaks?”) and intensity (“what each family says?”) – and to observe their respective contributions in time. I think this separation between those two mechanisms, applied on the transition between consecutive elections rather than globally, is the part of the analysis that would benefit the most from further work.

Acknowledgements

I would like to stress again that the modelling infrastructure was inspired by the topic-modelling pipeline I worked on at the European Central Bank in 2025 : this is where I learnt how to preprocess thoroughly textual data in preparation of topic modelling models, how to code them with Abstract classes in Python and how to finetune the hyperparameters of such models. I would like also to thank you, Mister Kermovant, for making available to us such an interesting corpus to work on! And for the record, I used a generative AI model (Claude) for help with (i) the descriptive statistics functions (because I already know how to code this), (ii) the plotting functions to get the charts aligned with the NEURIPS formatting guidelines (because it is not so interesting to code), (iii) understanding OCR errors and how to enhance the preprocessing accordingly (I had never worked with OCRed textual data before), (iv) with normalizing the different wording of candidates political support and professions (to reduce the number of different values and to be able to classify those into historically sensible and meaningful categories) and finally (v) correcting a bit my English for this report. Most importantly: all preprocessing ideas, design and modelling choices, all the code, results and interpretations are mine.

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A Temporal analysis of the socio-demographic characteristics of the candidates

A.1 Is there some feminisation or aging or youthening of the candidacies?

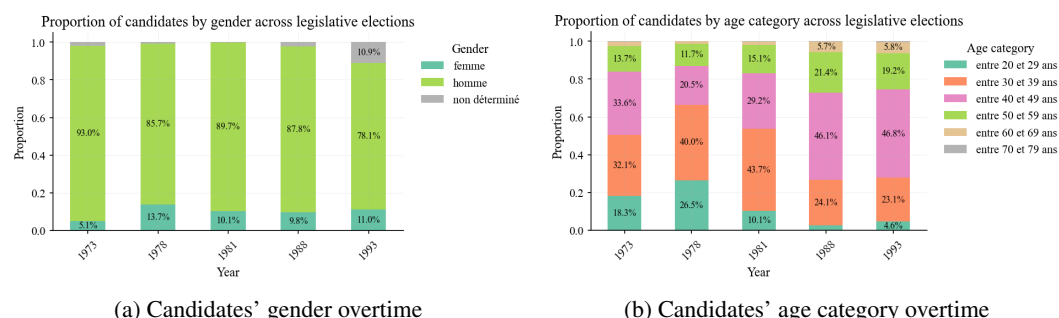


Figure 10: Is there some feminisation or aging or youthening of the candidacies?

Conclusion. It seems that there is no feminisation of the candidacies with a share of female candidacies stable around 10% (except if the "non déterminé" category of 1993 election is actually an artefact and represent mainly female candidates, but there was no sign in this direction). Regarding the age of the candidates, it seems that there is an aging phenomenon, consistent with sociological facts of the time (professionalisation of politics, extension of the duration of studies, etc.) with the share of younger than 29 years old almost disappearing, the share of older than 60 years old appearing, the share of 30-49 years old dropping by 10 percentage points, and the share of 40-49 years old (respectively 50-59 years old) increasing to reach almost 50% (almost 20%) of the candidacies.

A.2 Are some professions over-represented among the candidates, with what evolution?

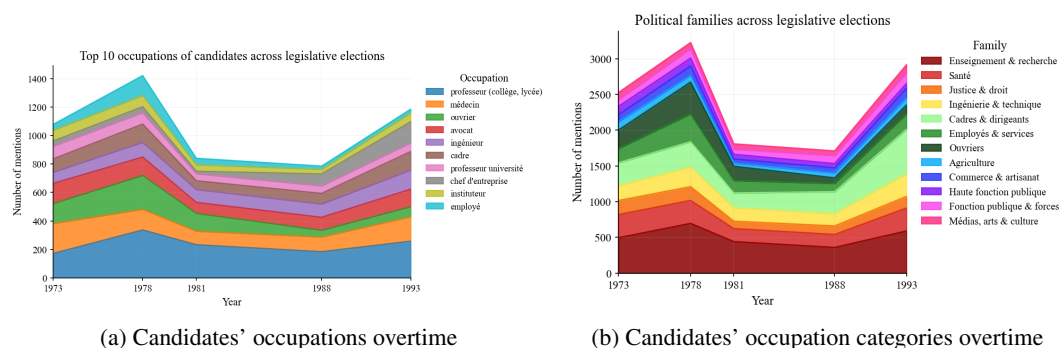


Figure 11: How do the most represented professions among candidates evolve over time?

Conclusion. The distribution of candidates' occupations is heavily skewed towards three intellectual professions (school teachers, doctors, and lawyers) which together dominate the top 10 across all elections, a classic feature of Western political recruitment where legal literacy and social capital are decisive. Workers (*ouvriers*) do appear in the ranking, but already in the 1970s their share is well below their weight in the active French population (roughly a third of the workforce at the time), meaning that blue-collar candidates were structurally under-represented even at their peak. The dynamics over time closely track the evolution of French society: the share of workers shrinks drastically, and the same holds for *employés*, reflecting deindustrialization and tertiarization, while engineers, *cadres*, and business leaders gain ground — a pattern made even clearer in the second chart, where the "Cadres & dirigeants" and "Ingénierie & technique" families expand at the expense of "Ouvriers" and "Employés & services". A few additional remarks: the "Enseignement & recherche" and "Santé" families remain remarkably stable across the whole period, confirming that intellectual professions form the durable backbone of French political personnel, and the sharp drop in total candidates in 1981 likely reflects the snap election following Mitterrand's victory rather than a sociological shift.

A.3 Cross-sectional study of the political support and occupations of the candidates

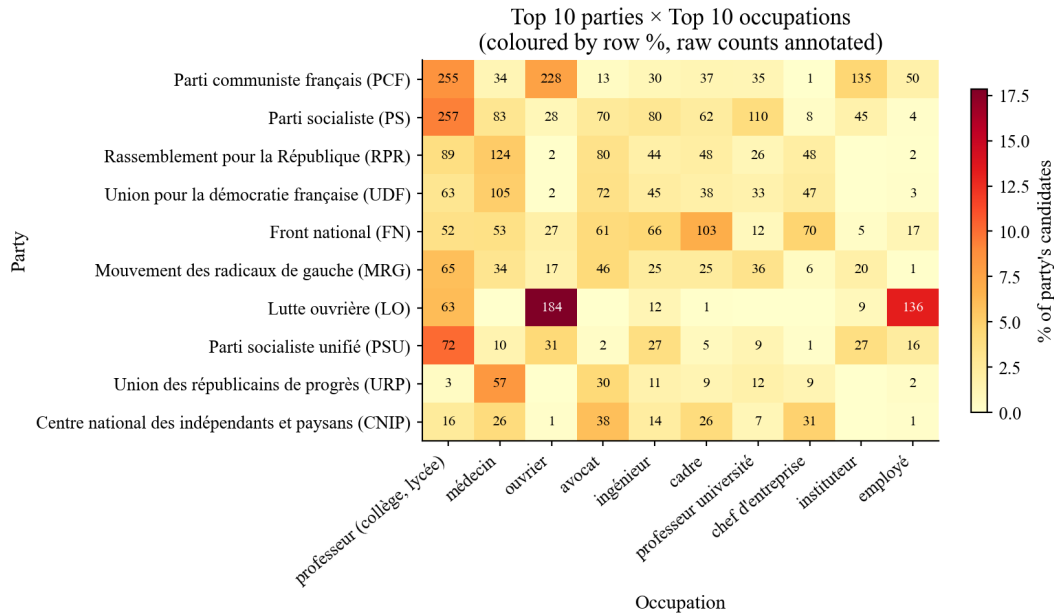


Figure 12: Cross-table of the occupations of the candidates of the top 10 parties

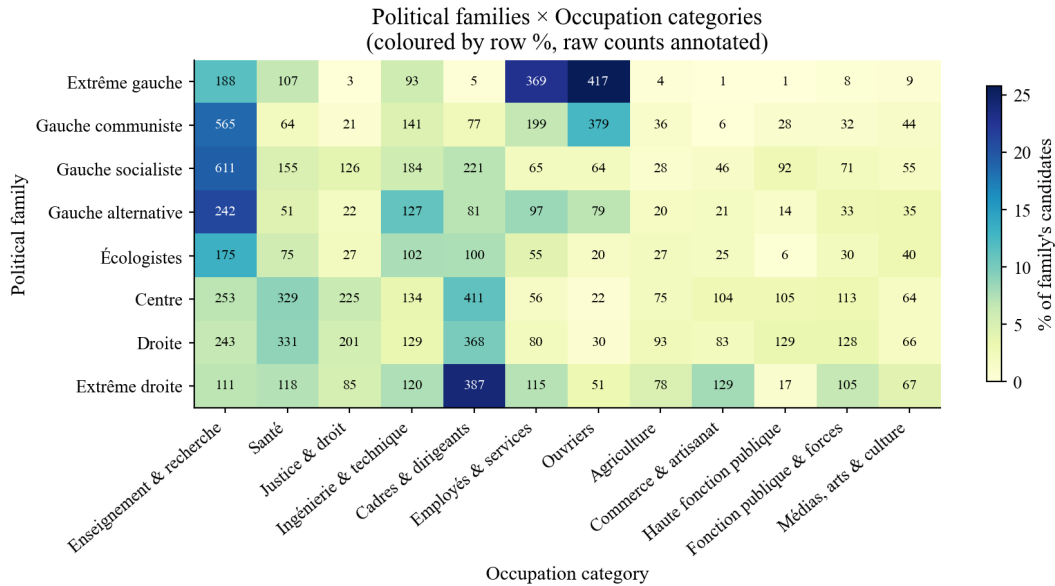


Figure 13: Cross-table of the occupations categories of the candidates of the 8 political families

B Hierarchical clustering of BERTopic topics

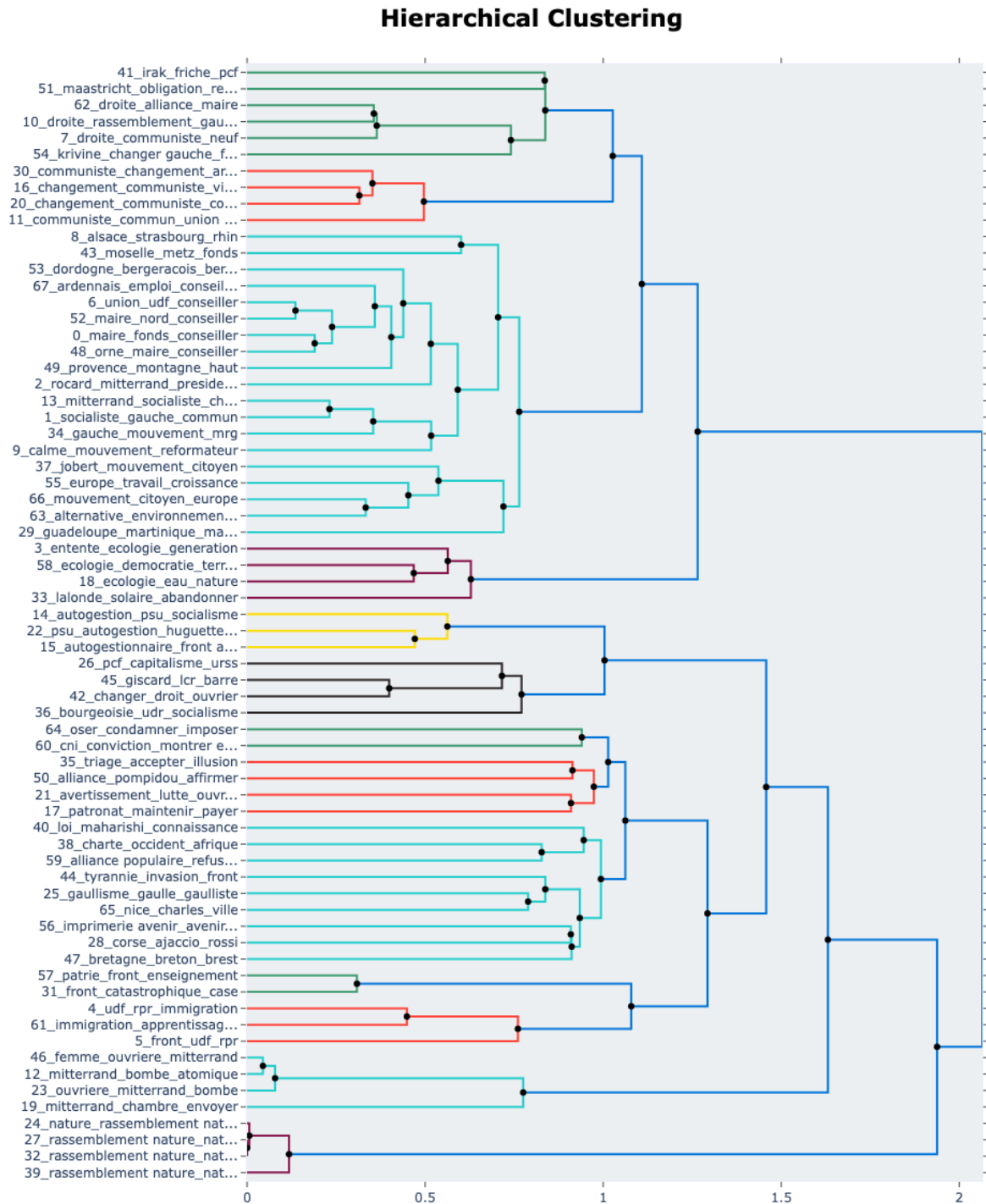


Figure 14: Hierarchical clustering of BERT topics, starting from the 68 ones

C Topics inferred by LDA and BERTopic

LDA — Word clouds



Figure 15: LDA 9 topics Word Clouds

BERTopic — word clouds

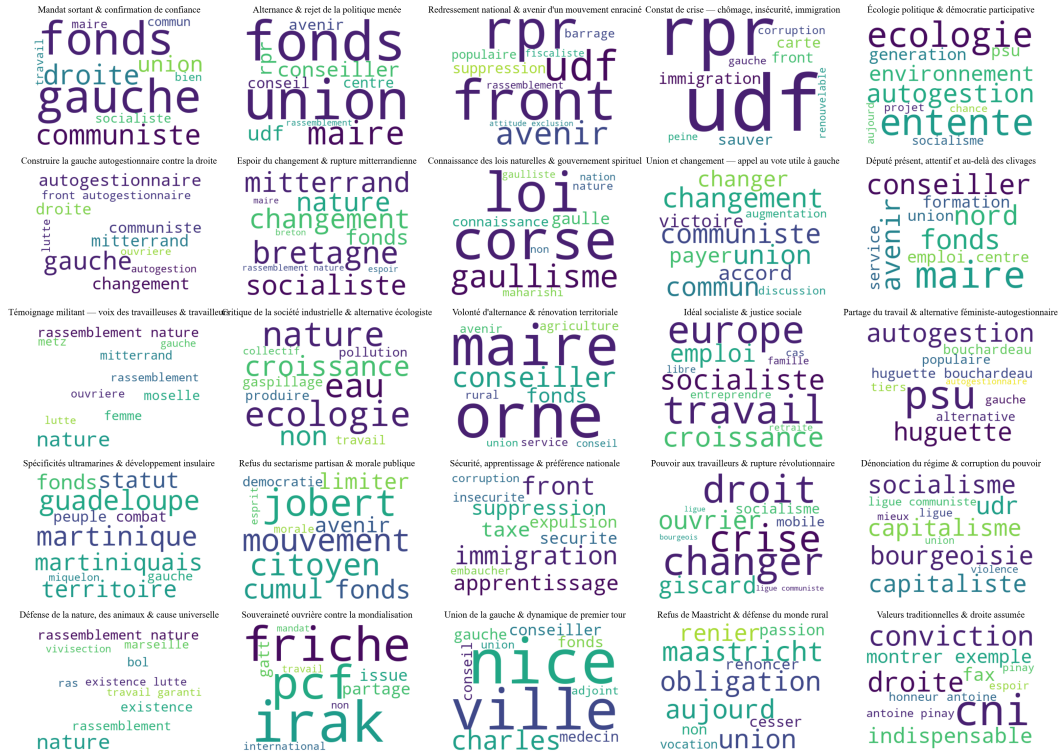


Figure 16: BERTopic 14 topics Word Clouds

D Results: topics by parties & occupations, evolving across time, the reciprocal analysis as with BERTopic but with NMF

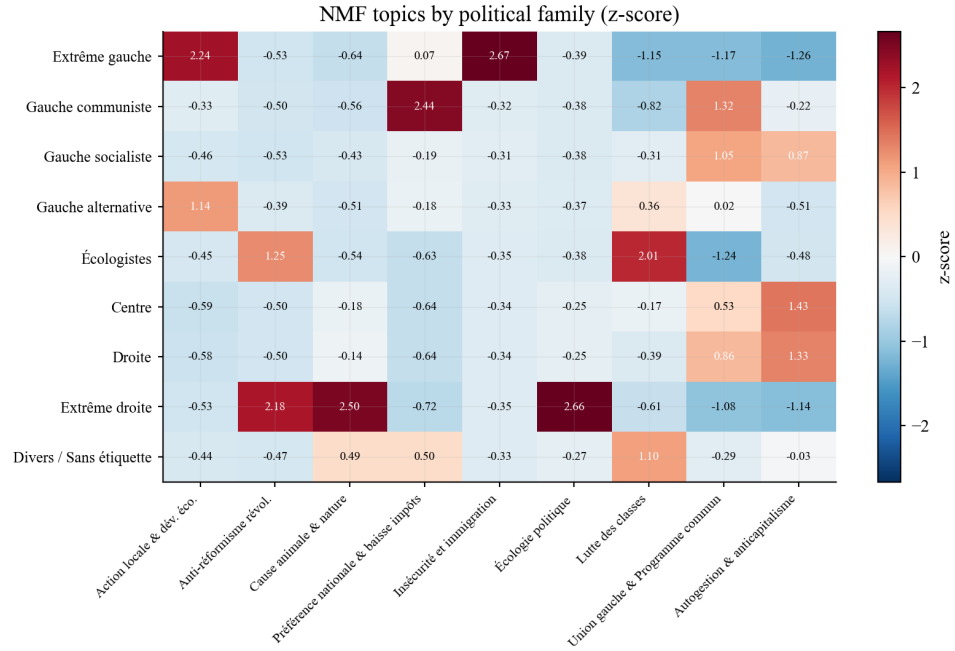


Figure 17: NMF topics by political family, expressed as a per-party z-score (red: over-represented, blue: under-represented)

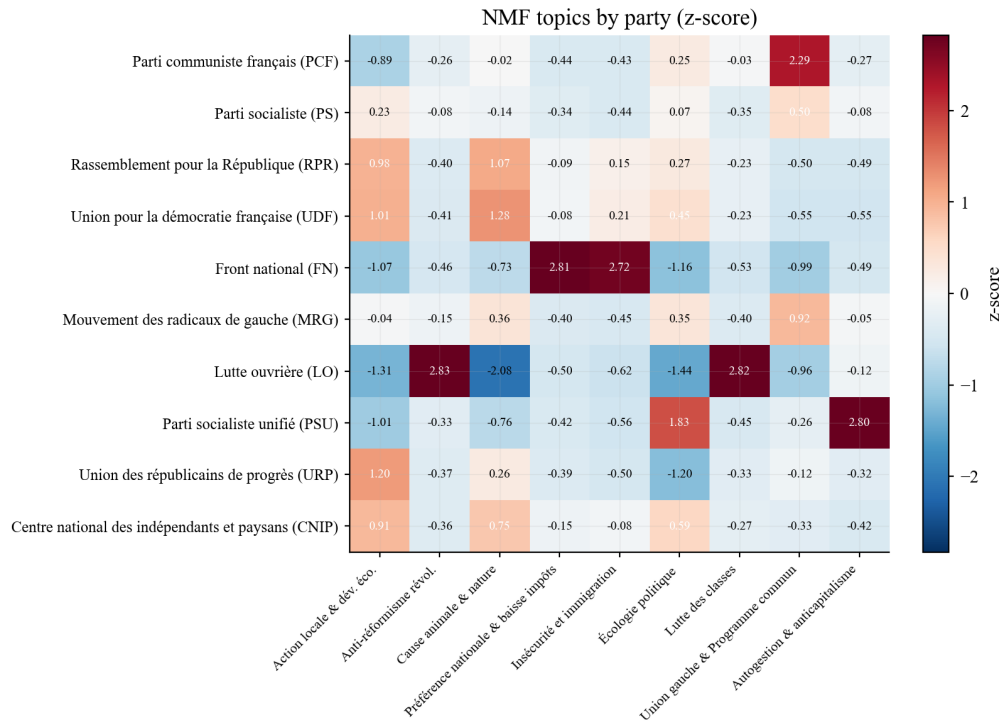


Figure 18: NMF topics of the top 10 parties, expressed as a per-party z-score (red: over-represented, blue: under-represented)

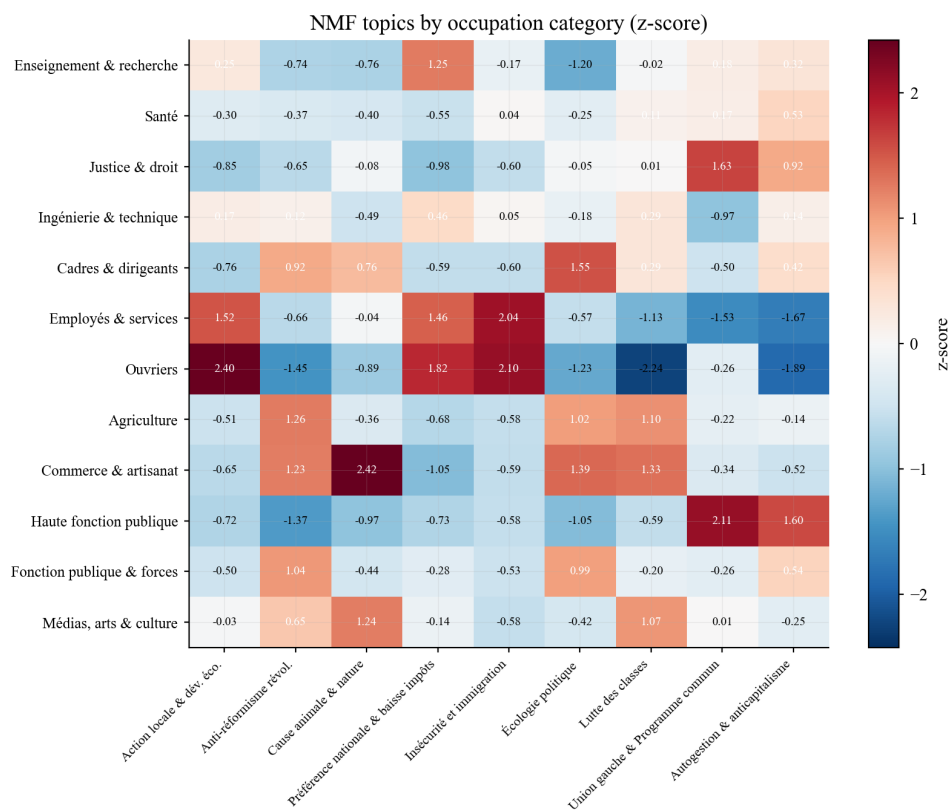


Figure 19: NMF topics by occupation category, expressed as a per-party z-score (red: over-represented, blue: under-represented)

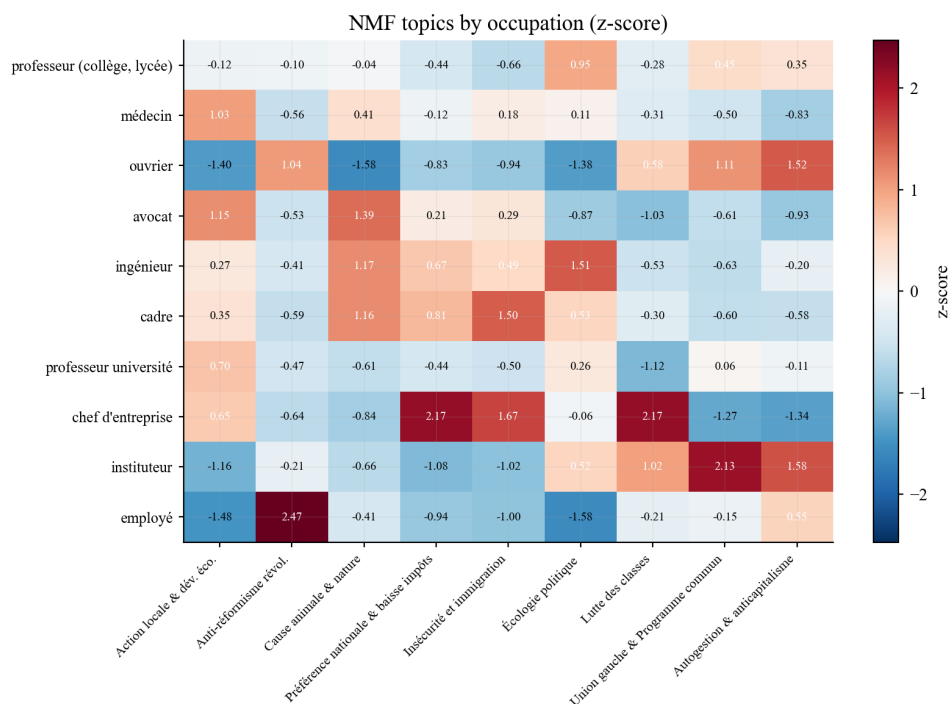


Figure 20: NMF topics of the top 10 occupations, expressed as a per-party z-score (red: over-represented, blue: under-represented)

E Results: topics by parties & occupations, evolving across time, more analysis with BERTopic

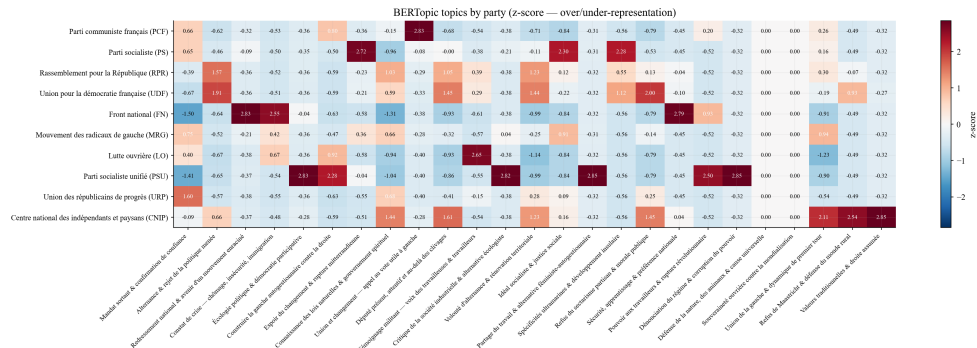


Figure 21: BERTopic topics of the top 10 parties, expressed as a per-party z-score (red: over-represented, blue: under-represented)

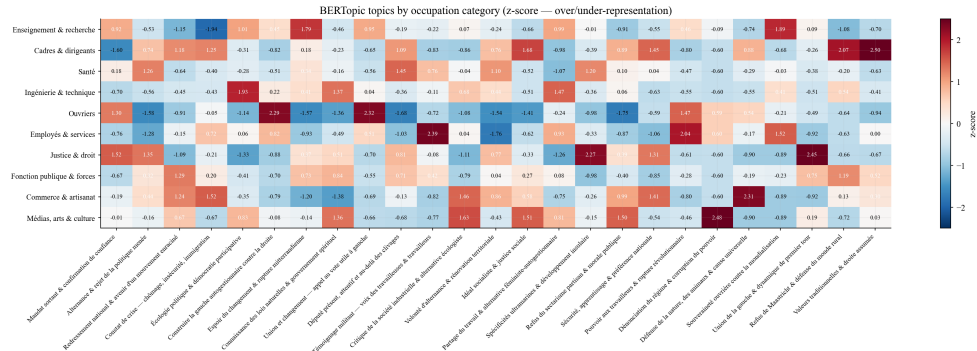


Figure 22: BERTopic topics by occupation category, expressed as a per-party z-score (red: over-represented, blue: under-represented)

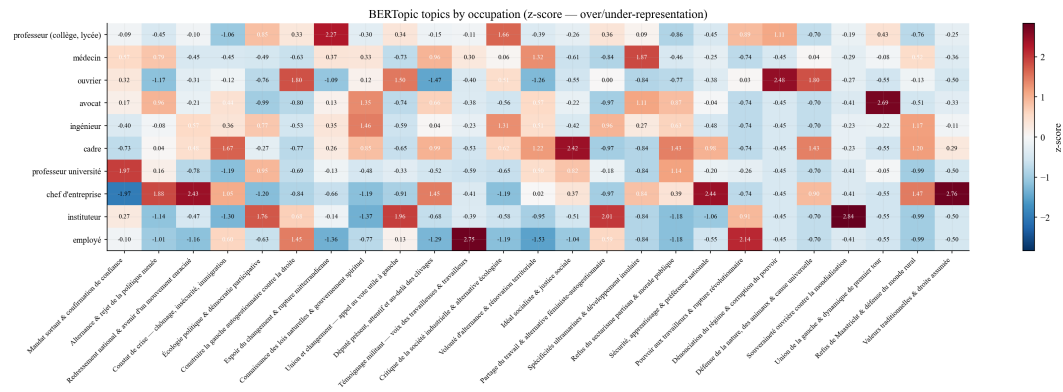


Figure 23: BERTopic topics of the top 10 occupations, expressed as a per-party z-score (red: over-represented, blue: under-represented)

F Temporal evolution of the topics with LDA and NMF

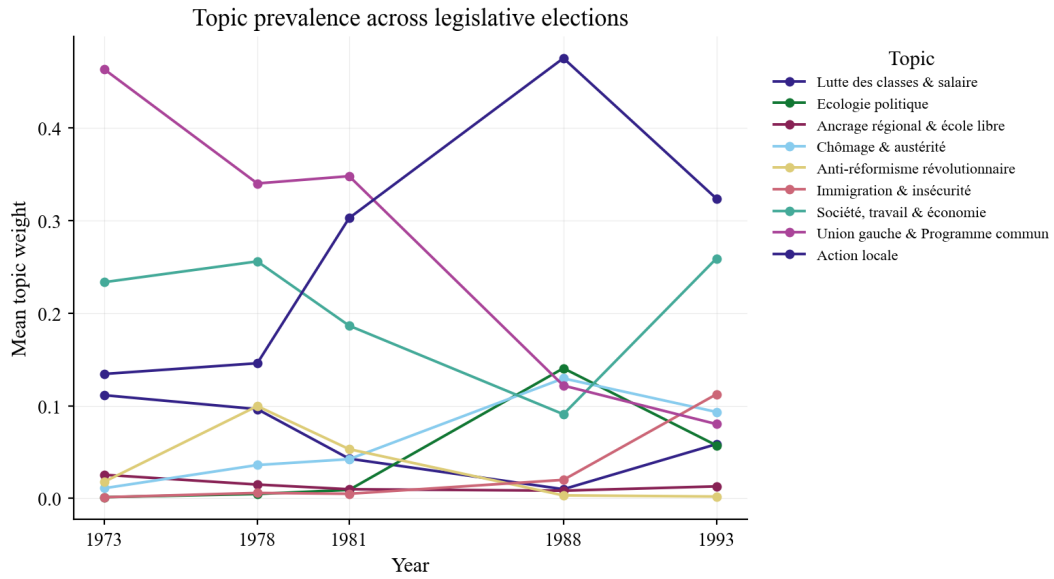


Figure 24: Caption

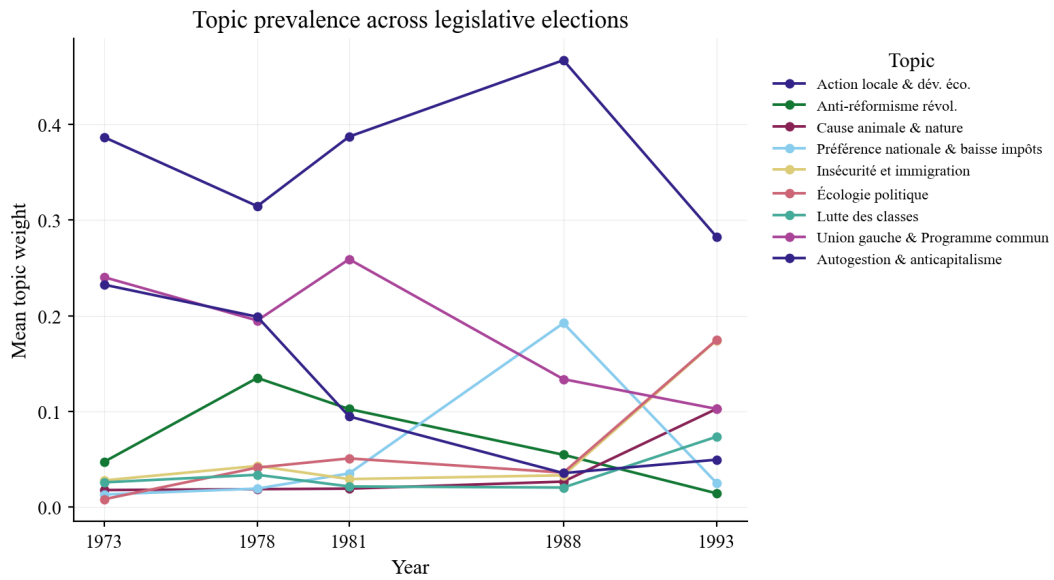


Figure 25: Caption

G Shift-share decomposition via LDA and NMF, and on the overall period for the 3 models

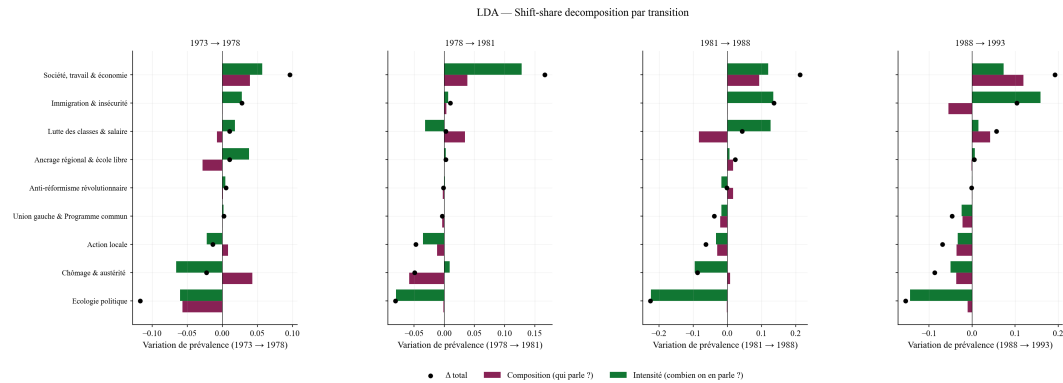


Figure 26: Shift-share decomposition by elections based on LDA topics

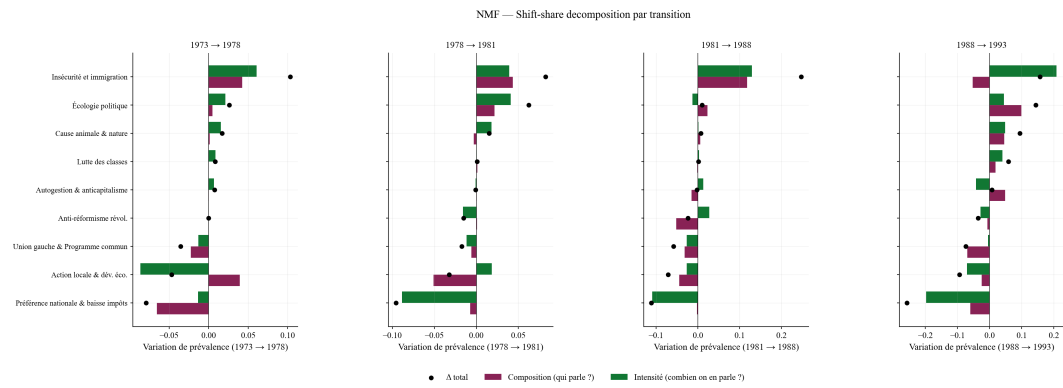


Figure 27: Shift-share decomposition by elections based on NMF topics

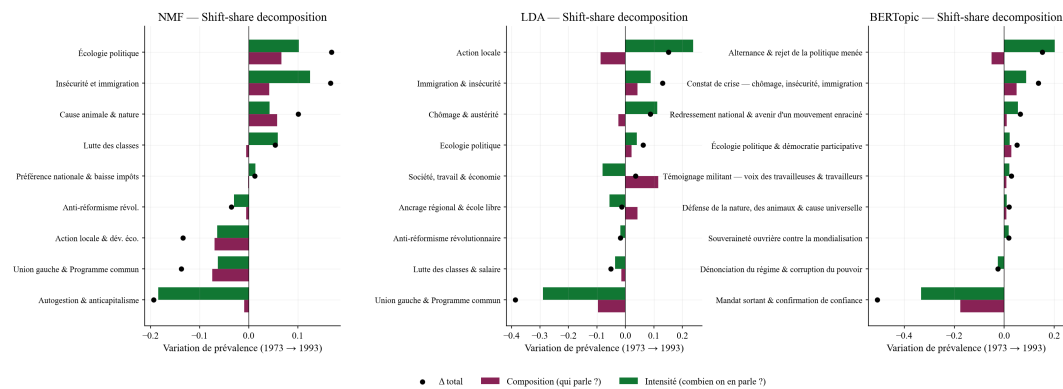


Figure 28: Shift-share decomposition on the 20 years period (1973 – 1993) for the 3 models